

for mobile and automotive platforms, highlighting the importance of on-device processing in reducing latency and improving performance.

An interesting trend report by Gartner suggests that by 2026, as much as 30% of workers will utilise digital "charisma filters" to enhance their professional interactions and career progression. These filters, powered by on-device generative AI, suggest a trend toward embedding

AI functionalities within personal devices to improve user experiences. Let me tell you, Gartner isn't alone in this assessment.

Deloitte predicts that by the end of 2025, over 30% of smartphones shipped will be equipped with location generative AI capabilities. This integration is expected to transform user interactions and drive substantial growth in the mobile device market. In a similar vein, IDC forecasts a 73.1%

increase in generative AI capable smartphone shipments in 2025, followed by consistent double-digit growth in subsequent years. On the PC front, according to Canalys, AI-capable PC shipments are projected to surpass 100 million units in 2025, accounting for 40% of all PC shipments. This indicates a substantial shift towards integrating AI functionalities directly into personal computing and mobile devices going forward.

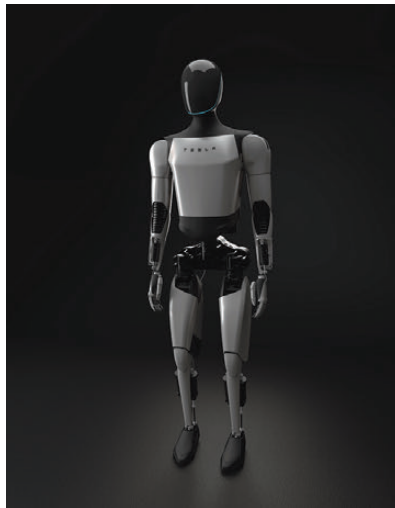
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MAKE WAY FOR AI-POWERED HUMANOID BOTS

The year 2025 is poised to be a pivotal milestone in the evolution of AI-powered humanoid robot systems, marking significant advancements in their development and integration across various sectors. According to a report by Goldman Sachs, the global market for humanoid robots is projected to reach \$38 billion by 2035, with substantial growth anticipated to commence around 2025. This surge is attributed to rapid advancements in artificial intelligence, particularly in large language models and end-to-end AI systems, which enhance the capabilities and versatility of humanoid robots.

A company called Figure AI is focused on creating general-purpose humanoid robots capable of performing a variety of tasks. Their latest model, Figure 02, unveiled in August 2024, features a sleek design with integrated cabling, enhanced battery capacity, and advanced AI capabilities for conversational interactions. The robot is currently being tested in BMW's manufacturing facilities.

Similarly, Clone Robotics, a Polish company, has developed a humanoid robot with water-powered muscles that mimic human movements and strength. 1X is a Norwegian company developing humanoid robots to assist



humans in everyday tasks. Their robot, EVE, is designed for roles in security, healthcare, and other service-oriented industries, focusing on safety and affordability. Chinese electric vehicle manufacturer Xpeng has introduced the "Iron" robot, a six-foot-tall humanoid designed with advanced AI and mechanical capabilities. This development is part of Xpeng's broader strategy to integrate robotics into their product lineup, including innovations like a new hybrid system and a modular flying car.

Elon Musk, CEO of Tesla, has been vocal about the company's ambi-

tions in humanoid robotics. In July 2024, Musk announced that Tesla's humanoid robot, Optimus, would enter low-volume production for internal use by early 2025, with plans for broader deployment later in the year through 2026. He emphasised that these robots are designed to perform tasks that are "boring, repetitive, and dangerous," aiming to enhance productivity and safety in various environments. Musk envisions a future where humanoid robots are commonplace, assisting in both industrial and domestic settings.

Similarly, Jensen Huang, CEO of NVIDIA, has expressed a strong commitment to advancing AI and robotics technologies. At NVIDIA's GTC 2024 conference, Huang discussed the company's efforts in developing AI platforms that support humanoid robots. He introduced the Jetson Thor computer, designed to enhance the capabilities of humanoid robots by providing advanced AI processing power. This technology aims to improve robots' autonomy and their ability to interact seamlessly with human environments.

Huang has emphasised the importance of creating AI models that enable robots to understand and navigate complex human settings. He highlighted NVIDIA's work on simulation platforms that allow for the training and testing of robots in virtual environments before deployment in the real world. This approach is intended to accelerate the development of safe and efficient humanoid robots.